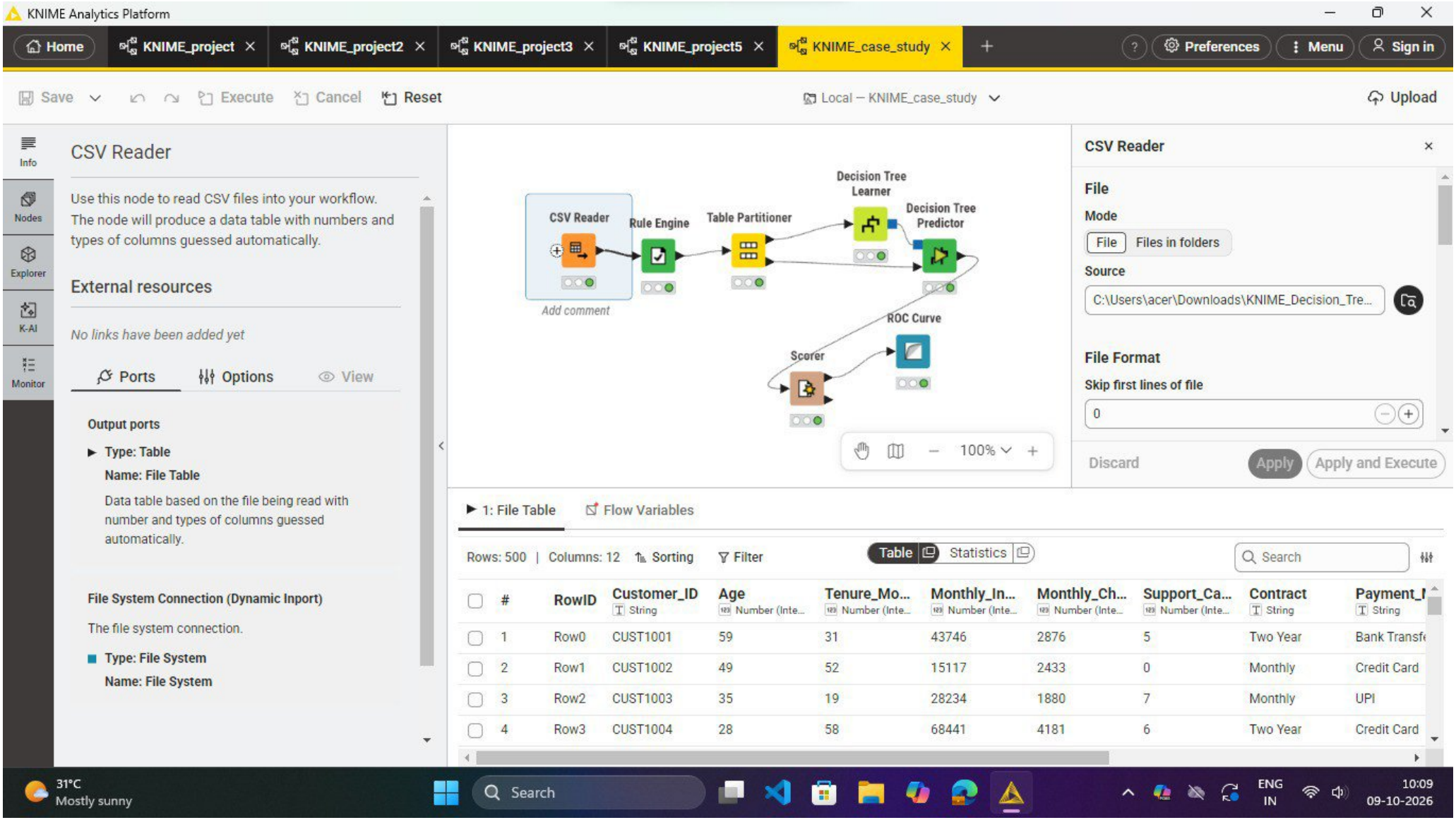
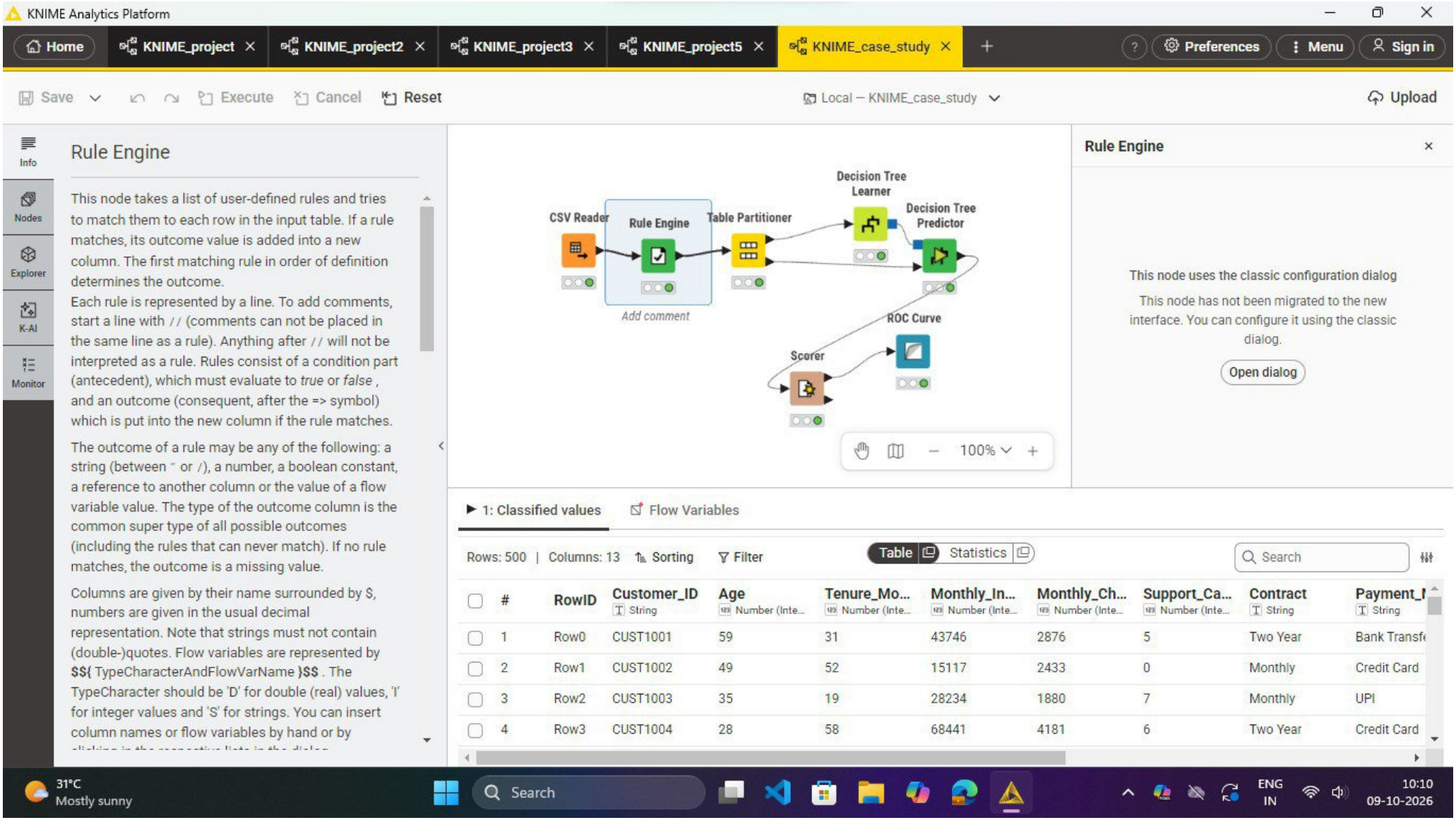
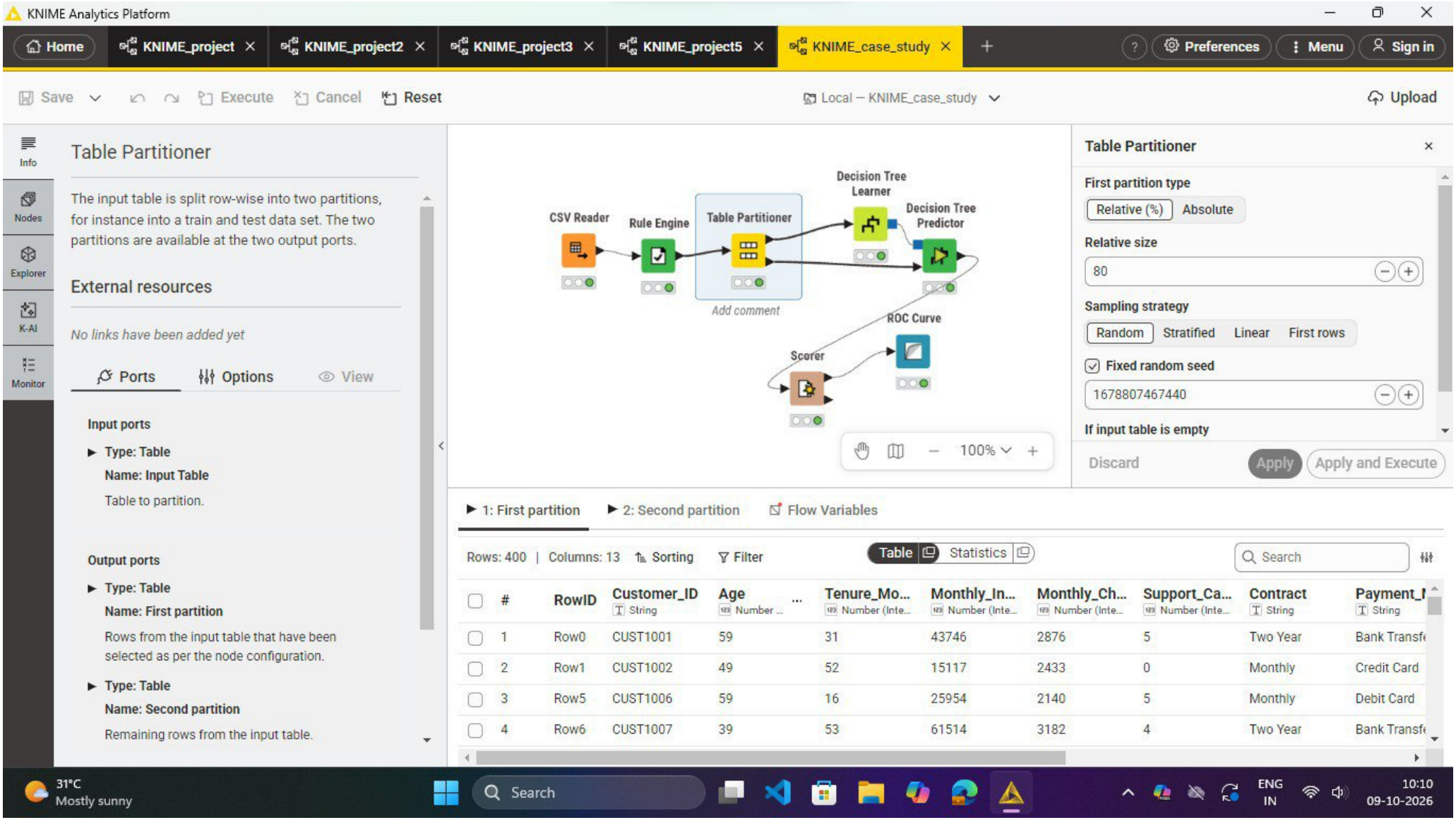
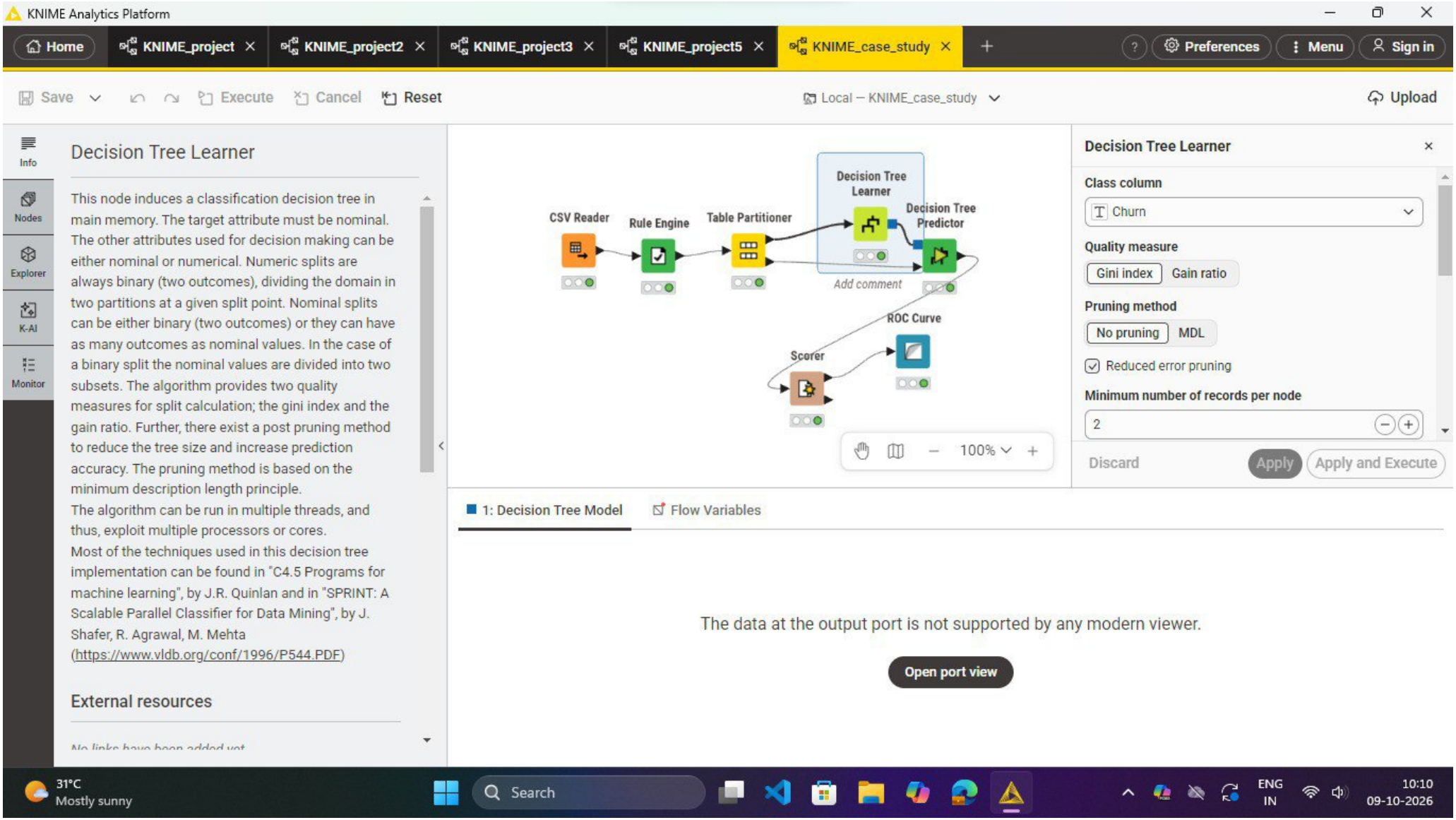
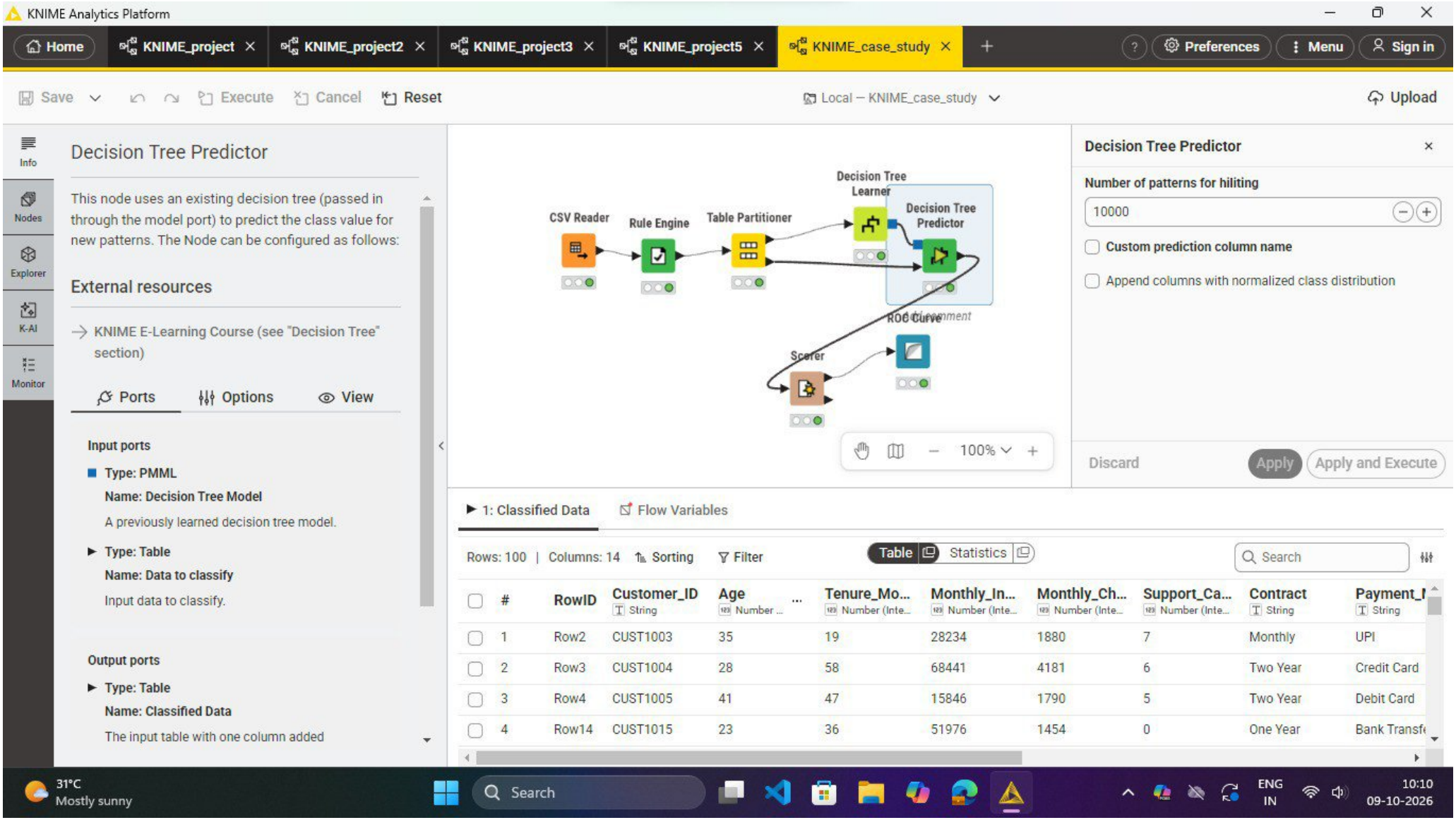
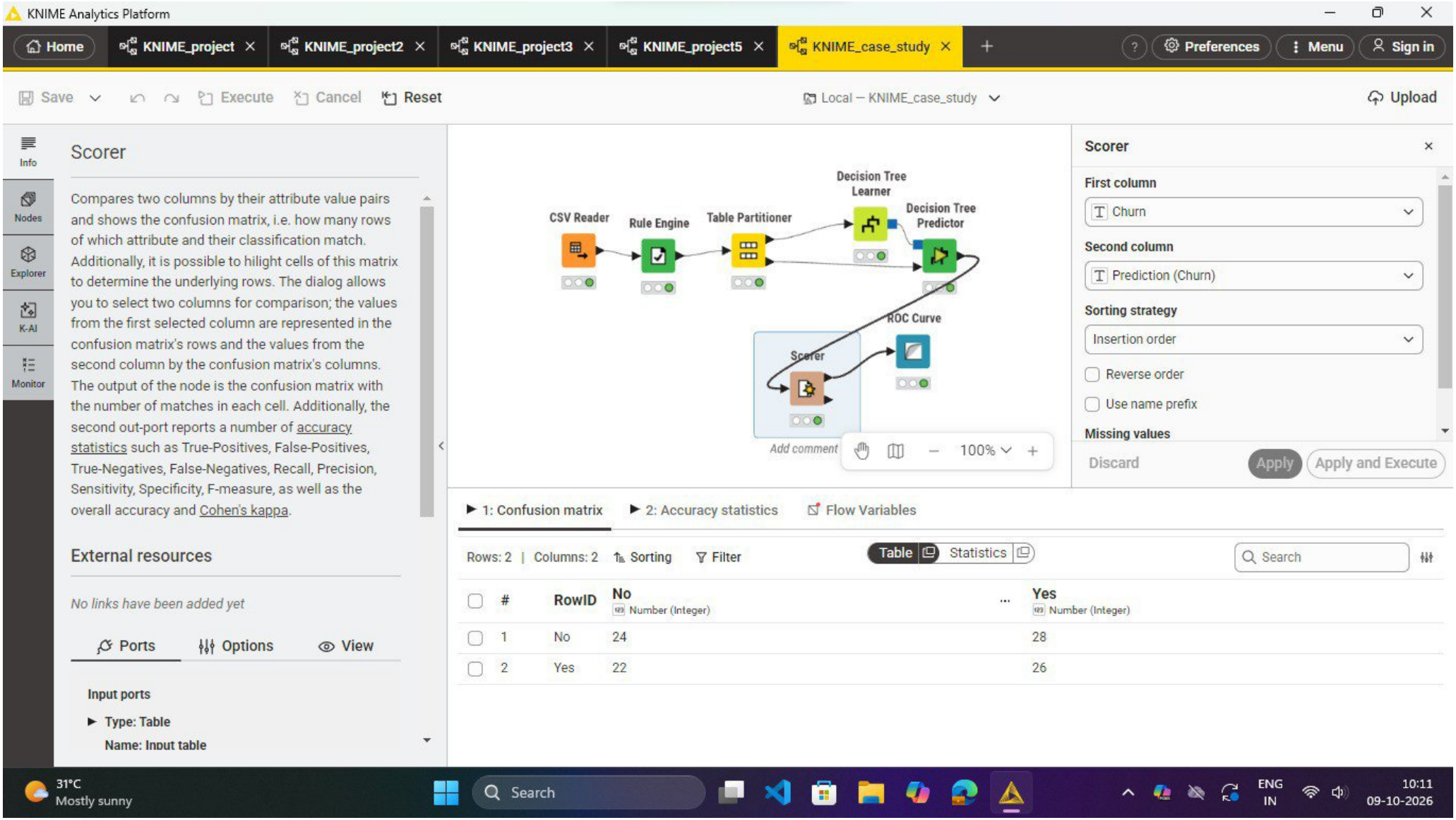

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CSV Reader
Use this node to read CSV files into your workflow. The node will produce a data table with numbers and types of columns guessed automatically.
External resources
No links have been added yet
Ports
Options
View
Output ports
Type: Table
Name: File Table
Data table based on the file being read with number and types of columns guessed automatically.
File System Connection (Dynamic Inport)
The file system connection.
Type: File System
Name: File System
CSV Reader
Rule Engine
Table Partitioner
Decision Tree Learner
Decision Tree Predictor
ROC Curve
Scorer
Add comment
100%
CSV Reader
File
Mode
File
Files in folders
Source
C:\Users\acer\Downloads\KNIME_Decision_Tre...
File Format
Skip first lines of file
0
Discard
Apply
Apply and Execute
1: File Table
Flow Variables
Rows: 500 | Columns: 12
Sorting
Filter
Table
Statistics
Search
RowID Customer_ID Age Tenure_Mo... Monthly_In... Monthly_Ch... Support_Ca... Contract Payment_I
1 Row0 CUST1001 59 31 43746 2876 5 Two Year Bank Transf
2 Row1 CUST1002 49 52 15117 2433 0 Monthly Credit Card
3 Row2 CUST1003 35 19 28234 1880 7 Monthly UPI
4 Row3 CUST1004 28 58 68441 4181 6 Two Year Credit Card
31°C
Mostly sunny
Search
ENG
IN
10:09
09-10-2026

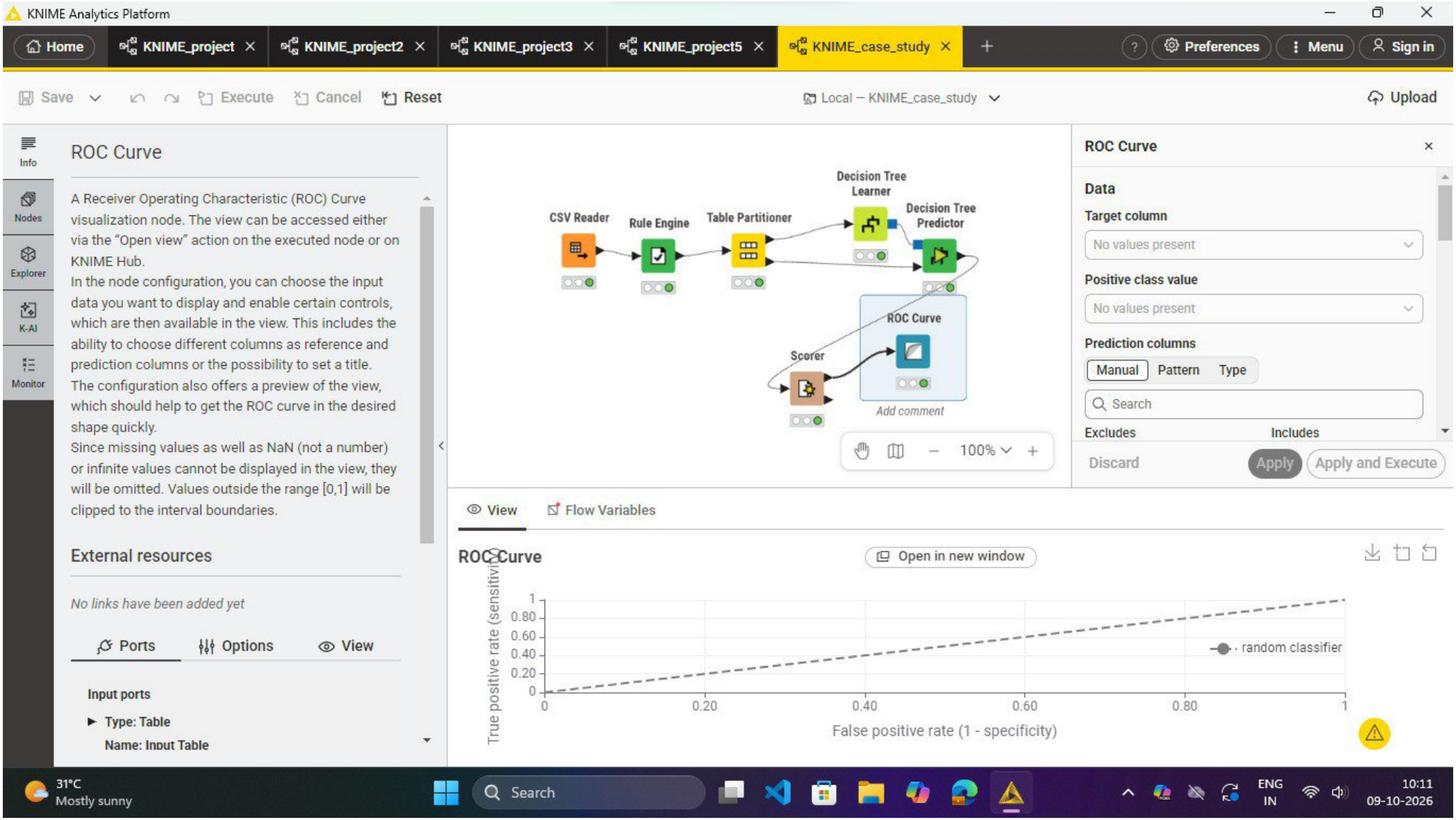

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Rule Engine
This node takes a list of user-defined rules and tries to match them to each row in the input table. If a rule matches, its outcome value is added into a new column. The first matching rule in order of definition determines the outcome.
Each rule is represented by a line. To add comments, start a line with // (comments can not be placed in the same line as a rule). Anything after // will not be interpreted as a rule. Rules consist of a condition part (antecedent), which must evaluate to true or false , and an outcome (consequent, after the => symbol) which is put into the new column if the rule matches.
The outcome of a rule may be any of the following: a string (between " or /), a number, a boolean constant, a reference to another column or the value of a flow variable value. The type of the outcome column is the common super type of all possible outcomes (including the rules that can never match). If no rule matches, the outcome is a missing value.
Columns are given by their name surrounded by $, numbers are given in the usual decimal representation. Note that strings must not contain (double-)quotes. Flow variables are represented by $${ TypeCharacterAndFlowVarName }$$. The TypeCharacter should be 'D' for double (real) values, 'I' for integer values and 'S' for strings. You can insert column names or flow variables by hand or by
CSV Reader
Rule Engine
Table Partitioner
Decision Tree Learner
Decision Tree Predictor
ROC Curve
Scorer
Add comment
100%
Rule Engine
This node uses the classic configuration dialog
This node has not been migrated to the new interface. You can configure it using the classic dialog.
Open dialog
1: Classified values
Flow Variables
Rows: 500 | Columns: 13
Sorting
Filter
Table
Statistics
Search
RowID Customer_ID String Age Number (Inte... Tenure_Mo... Number (Inte... Monthly_In... Number (Inte... Monthly_Ch... Number (Inte... Support_Ca... Number (Inte... Contract String Payment_I String
1 Row0 CUST1001 59 31 43746 2876 5 Two Year Bank Transf
2 Row1 CUST1002 49 52 15117 2433 0 Monthly Credit Card
3 Row2 CUST1003 35 19 28234 1880 7 Monthly UPI
4 Row3 CUST1004 28 58 68441 4181 6 Two Year Credit Card
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Table Partitioner
The input table is split row-wise into two partitions, for instance into a train and test data set. The two partitions are available at the two output ports.
External resources
No links have been added yet
Ports
Options
View
Input ports
Type: Table
Name: Input Table
Table to partition.
Output ports
Type: Table
Name: First partition
Rows from the input table that have been selected as per the node configuration.
Type: Table
Name: Second partition
Remaining rows from the input table.
CSV Reader
Rule Engine
Table Partitioner
Decision Tree Learner
Decision Tree Predictor
ROC Curve
Scorer
Add comment
100%
Table Partitioner
First partition type
Relative (%)
Absolute
Relative size
80
Sampling strategy
Random
Stratified
Linear
First rows
Fixed random seed
1678807467440
If input table is empty
Discard
Apply
Apply and Execute
1: First partition
2: Second partition
Flow Variables
Rows: 400 | Columns: 13
Sorting
Filter
Table
Statistics
Search
RowID Customer_ID Age Tenure_Mo... Monthly_In... Monthly_Ch... Support_Ca... Contract Payment_I
1 Row0 CUST1001 59 31 43746 2876 5 Two Year Bank Transfe
2 Row1 CUST1002 49 52 15117 2433 0 Monthly Credit Card
3 Row5 CUST1006 59 16 25954 2140 5 Monthly Debit Card
4 Row6 CUST1007 39 53 61514 3182 4 Two Year Bank Transfe
31°C
Mostly sunny
Search
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Decision Tree Learner
This node induces a classification decision tree in main memory. The target attribute must be nominal. The other attributes used for decision making can be either nominal or numerical. Numeric splits are always binary (two outcomes), dividing the domain in two partitions at a given split point. Nominal splits can be either binary (two outcomes) or they can have as many outcomes as nominal values. In the case of a binary split the nominal values are divided into two subsets. The algorithm provides two quality measures for split calculation; the gini index and the gain ratio. Further, there exist a post pruning method to reduce the tree size and increase prediction accuracy. The pruning method is based on the minimum description length principle.
The algorithm can be run in multiple threads, and thus, exploit multiple processors or cores.
Most of the techniques used in this decision tree implementation can be found in "C4.5 Programs for machine learning", by J.R. Quinlan and in "SPRINT: A Scalable Parallel Classifier for Data Mining", by J. Shafer, R. Agrawal, M. Mehta (https://www.vldb.org/conf/1996/P544.PDF)
External resources
CSV Reader
Rule Engine
Table Partitioner
Decision Tree Learner
Decision Tree Predictor
Add comment
ROC Curve
Scorer
100%
Decision Tree Learner
Class column
Churn
Quality measure
Gini index
Gain ratio
Pruning method
No pruning
MDL
Reduced error pruning
Minimum number of records per node
2
Discard
Apply
Apply and Execute
1: Decision Tree Model
Flow Variables
The data at the output port is not supported by any modern viewer.
Open port view
31°C
Mostly sunny
Search
ENG
IN
10:10
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K-AI
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Decision Tree Predictor
This node uses an existing decision tree (passed in through the model port) to predict the class value for new patterns. The Node can be configured as follows:
External resources
KNIME E-Learning Course (see "Decision Tree" section)
Ports
Options
View
Input ports
Type: PMML
Name: Decision Tree Model
A previously learned decision tree model.
Type: Table
Name: Data to classify
Input data to classify.
Output ports
Type: Table
Name: Classified Data
The input table with one column added
CSV Reader
Rule Engine
Table Partitioner
Decision Tree Learner
Decision Tree Predictor
ROC Curve
Scorer
100%
Decision Tree Predictor
Number of patterns for hiliting
10000
Custom prediction column name
Append columns with normalized class distribution
Discard
Apply
Apply and Execute
1: Classified Data
Flow Variables
Rows: 100 | Columns: 14
Sorting
Filter
Table
Statistics
Search
RowID Customer_ID Age Tenure_Mo... Monthly_In... Monthly_Ch... Support_Ca... Contract Payment_I
1 Row2 CUST1003 35 19 28234 1880 7 Monthly UPI
2 Row3 CUST1004 28 58 68441 4181 6 Two Year Credit Card
3 Row4 CUST1005 41 47 15846 1790 5 Two Year Debit Card
4 Row14 CUST1015 23 36 51976 1454 0 One Year Bank Transf
31°C
Mostly sunny
Search
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Compares two columns by their attribute value pairs and shows the confusion matrix, i.e. how many rows of which attribute and their classification match. Additionally, it is possible to hilight cells of this matrix to determine the underlying rows. The dialog allows you to select two columns for comparison; the values from the first selected column are represented in the confusion matrix's rows and the values from the second column by the confusion matrix's columns. The output of the node is the confusion matrix with the number of matches in each cell. Additionally, the second out-port reports a number of accuracy statistics such as True-Positives, False-Positives, True-Negatives, False-Negatives, Recall, Precision, Sensitivity, Specificity, F-measure, as well as the overall accuracy and Cohen's kappa.
External resources
No links have been added yet
Ports
Options
View
Input ports
Type: Table
Name: Input table
CSV Reader
Rule Engine
Table Partitioner
Decision Tree Learner
Decision Tree Predictor
ROC Curve
Scorer
Add comment
100%
Scorer
First column
Churn
Second column
Prediction (Churn)
Sorting strategy
Insertion order
Reverse order
Use name prefix
Missing values
Discard
Apply
Apply and Execute
1: Confusion matrix
2: Accuracy statistics
Flow Variables
Rows: 2 | Columns: 2
Sorting
Filter
Table
Statistics
Search
RowID No Number (Integer) ... Yes Number (Integer)
1 No 24 28
2 Yes 22 26
31°C
Mostly sunny
Search
ENG
IN
10:11
09-10-2026


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ROC Curve
A Receiver Operating Characteristic (ROC) Curve visualization node. The view can be accessed either via the "Open view" action on the executed node or on KNIME Hub.
In the node configuration, you can choose the input data you want to display and enable certain controls, which are then available in the view. This includes the ability to choose different columns as reference and prediction columns or the possibility to set a title.
The configuration also offers a preview of the view, which should help to get the ROC curve in the desired shape quickly.
Since missing values as well as NaN (not a number) or infinite values cannot be displayed in the view, they will be omitted. Values outside the range [0,1] will be clipped to the interval boundaries.
External resources
No links have been added yet
Ports
Options
View
Input ports
Type: Table
Name: Input Table
CSV Reader
Rule Engine
Table Partitioner
Decision Tree Learner
Decision Tree Predictor
ROC Curve
Scorer
Add comment
100%
ROC Curve
Data
Target column
No values present
Positive class value
No values present
Prediction columns
Manual
Pattern
Type
Search
Excludes
Includes
Discard
Apply
Apply and Execute
View
Flow Variables
ROC Curve
Open in new window
True positive rate (sensitivity)
False positive rate (1 - specificity)
random classifier
0
0.20
0.40
0.60
0.80
1
31°C
Mostly sunny
Search
ENG
IN
10:11
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